

Tools Required

- Python 3.x
- Jupyter Notebook
- NumPy
- Pandas
- Matplotlib / Seaborn

Step 1: Sample Dataset (Stock Returns)

	A	B	C	D	E	F
1	Day	AAPL	MSFT	GOOGL	AMZN	META
2	1	0.012	0.01	0.008	0.014	0.011
3	2	0.015	0.013	0.01	0.018	0.014
4	3	-0.008	-0.006	-0.012	-0.01	-0.009
5	4	0.02	0.018	0.015	0.022	0.019
6	5	0.005	0.006	0.004	0.007	0.006
7	6	-0.01	-0.009	-0.014	-0.012	-0.011
8	7	0.018	0.016	0.014	0.02	0.017
9	8	0.007	0.006	0.005	0.008	0.007
10	9	-0.004	-0.003	-0.006	-0.005	-0.004
11	10	0.011	0.01	0.009	0.012	0.01
12	11	0.013	0.011	0.01	0.015	0.012
13	12	-0.006	-0.005	-0.008	-0.007	-0.006
14	13	0.017	0.015	0.013	0.019	0.016
15	14	0.009	0.008	0.007	0.01	0.009
16	15	-0.005	-0.004	-0.006	-0.005	-0.004
17	16	0.014	0.012	0.011	0.016	0.013
18	17	0.006	0.005	0.004	0.007	0.006
19	18	-0.009	-0.008	-0.01	-0.009	-0.008
20	19	0.016	0.014	0.012	0.018	0.015
21	20	0.008	0.007	0.006	0.009	0.008
22	21	-0.007	-0.006	-0.009	-0.008	-0.007
23	22	0.019	0.017	0.015	0.021	0.018
24	23	0.01	0.009	0.008	0.011	0.01
25	24	-0.006	-0.005	-0.007	-0.006	-0.005
26	25	0.015	0.013	0.012	0.017	0.014
27	26	0.007	0.006	0.005	0.008	0.007
28	27	-0.008	-0.007	-0.01	-0.009	-0.008
29	28	0.018	0.016	0.014	0.02	0.017
30	29	0.009	0.008	0.007	0.01	0.009
31	30	0.013	0.012	0.01	0.015	0.013

Step 2: Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Step 3: Load Data

```
data = pd.read_excel("Stock data.xlsx")
returns = data.drop(columns=["Day"])
returns
```

Step 4: Descriptive Statistics

```
mean_values = returns.mean()
variance_values = returns.var()
mean_values
variance_values
```

Interpretation

- Mean represents average daily return of each stock
- Variance indicates volatility (risk level)

Step 5: Covariance Matrix

```
cov_matrix = returns.cov()
cov_matrix
```

Interpretation

- Positive covariance → stocks move in the same direction
- Higher covariance → higher portfolio risk
- Lower covariance → better diversification

Step 6: Correlation Matrix

```
corr_matrix = returns.corr()
Corr_matrix
```

Note

Correlation values range between -1 and $+1$.

Step 7: Visualization (Optional)

```
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix – Stock Returns")
plt.show()
```

Step 8: Interpretation

- Strong positive correlation exists among technology stocks
- Stocks influenced by similar market factors move together
- High correlation increases portfolio risk

Inference

- Diversification reduces unsystematic risk
- Selecting weakly correlated assets improves portfolio stability
- Covariance and correlation help in portfolio construction

OUTPUT:

```
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JupyterLab Python 3 (ipykernel)

[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

[2]: data = pd.read_excel("Stock data.xlsx")
data.head()

[2]:
```

	Day	AAPL	MSFT	GOOGL	AMZN	META
0	1	0.012	0.010	0.008	0.014	0.011
1	2	0.015	0.013	0.010	0.018	0.014
2	3	-0.008	-0.006	-0.012	-0.010	-0.009
3	4	0.020	0.018	0.015	0.022	0.019
4	5	0.005	0.006	0.004	0.007	0.006

```


[3]: returns = data.drop(columns=["Day"])
returns

[3]:
```

	AAPL	MSFT	GOOGL	AMZN	META
0	0.012	0.010	0.008	0.014	0.011
1	0.015	0.013	0.010	0.018	0.014
2	-0.008	-0.006	-0.012	-0.010	-0.009
3	0.020	0.018	0.015	0.022	0.019
4	0.005	0.006	0.004	0.007	0.006
5	-0.010	-0.009	-0.014	-0.012	-0.011
6	0.018	0.016	0.014	0.020	0.017
7	0.007	0.006	0.005	0.008	0.007
8	-0.004	-0.003	-0.006	-0.005	-0.004

```
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JupyterLab Python 3 (ipykernel) Trusted

[4]: returns_vector = returns.values
     returns_vector[:5]

[4]: array([[ 0.012,  0.01 ,  0.008,  0.014,  0.011],
           [ 0.015,  0.013,  0.01 ,  0.018,  0.014],
           [-0.008, -0.006, -0.012, -0.01 , -0.009],
           [ 0.02 ,  0.018,  0.015,  0.022,  0.019],
           [ 0.005,  0.006,  0.004,  0.007,  0.006]])

[5]: covariance_matrix = returns.cov()
     covariance_matrix

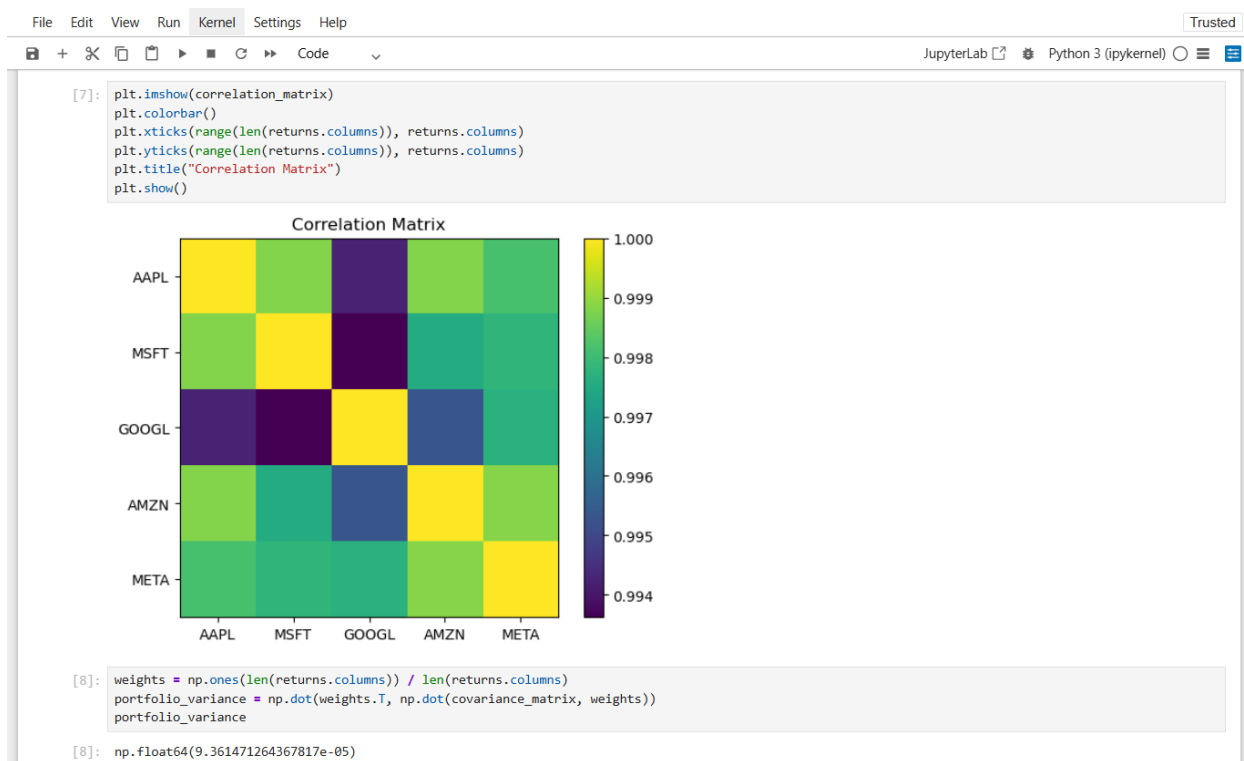
[5]:
```

	AAPL	MSFT	GOOGL	AMZN	META
AAPL	0.000098	0.000085	0.000091	0.000110	0.000094
MSFT	0.000085	0.000074	0.000079	0.000096	0.000082
GOOGL	0.000091	0.000079	0.000086	0.000103	0.000088
AMZN	0.000110	0.000096	0.000103	0.000124	0.000106
META	0.000094	0.000082	0.000088	0.000106	0.000090

```
[6]: correlation_matrix = returns.corr()
     correlation_matrix

[6]:
```

	AAPL	MSFT	GOOGL	AMZN	META
AAPL	1.000000	0.998808	0.994224	0.998807	0.998143
MSFT	0.998808	1.000000	0.993622	0.997542	0.997787
GOOGL	0.994224	0.993622	1.000000	0.995329	0.997700
AMZN	0.998807	0.997542	0.995329	1.000000	0.998840
META	0.998143	0.997787	0.997700	0.998840	1.000000



RUBRICS:

Problem Understanding (10)	Design Document (10)	Implementation (15)	Output (5)	Time Management (5)	Viva (5)	Total (50)

RESULT:

Thus, financial asset return data was successfully analyzed using descriptive statistics. Covariance and correlation matrices were computed to study relationships among stocks, helping to understand diversification benefits and portfolio risk exposure.